USE OF THE CONGRUENCE THEOREMS

How to read the figures. —— given - - - auxiliary (added in the proof)
—— what is being proved or asked

7–1 INTRODUCTION

In this chapter we do two things. In the first part we prove several inequalities involving sides and angles of a triangle, ending up with the *triangle inequality*. This enables us to prove that a line segment is the shortest path between two points. In the second part we prove that some of the constructions we made with ruler and compass in Chapter 1 are correct.

7–2 EXTERIOR ANGLES

In this section we prove a theorem about triangles which Euclid found to be very useful. We need a definition.

Definition 7–1. If an angle is adjacent and supplementary to an angle of a triangle, it is called an *exterior angle* of the triangle. The other two angles of the triangle are called the *opposite interior angles*. For example, $\angle DBC$ in Fig. 7–1 is an exterior angle and $\angle A$ and $\angle C$ are interior angles and opposite to $\angle DBC$.

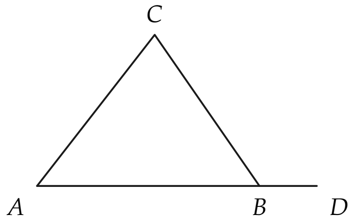


FIGURE 7–1

Exercise Group 7–1

1. In $\triangle ABC$, how many exterior angles are there having vertex B? How are these exterior angles related to $\angle B$?
2. How many exterior angles does a triangle have?
3. Do the words *opposite*, *interior*, and *exterior*, used in Definition 7–1, refer to properties or to relations?

Theorem 7–1. *An exterior angle of a triangle is greater than either opposite interior angle.*

We give two separate proofs of this theorem. The first proof may be found in Hilbert's work, the second in Euclid's *Elements*.
First Proof. We know that, for any two angles, the first angle is either $>$, or $<$, or $\cong$ the second. Therefore, we need only show that the exterior angle cannot be $\cong$ or $<$ an opposite interior angle, and it must follow that the exterior angle is $>$ an opposite interior angle. Suppose that the triangle is labeled as in Fig. 7–2.

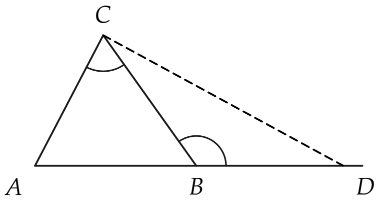


FIGURE 7–2

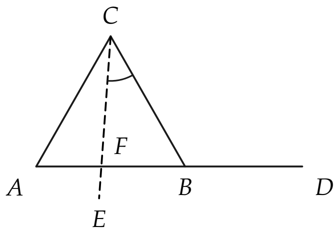


FIGURE 7–3

We first show that the exterior angle cannot be congruent to $\angle C$. Choose D, as in the figure, such that $BD \cong AC$, and suppose that $\angle DBC \cong \angle C$. Then $\triangle ACB \cong \triangle DBC$ because of the S.A.S. theorem. Hence $\angle CBA \cong \angle BCD$. Therefore, because of the addition of angles postulate, $\angle ACD \cong \angle ABD$. But $\angle ABD$ is a straight angle, and so $\angle ACD$ is a straight angle. This is impossible. (Why?) Hence $\angle DBC$ is *not* congruent to $\angle C$.

We now show that it is impossible that $\angle DBC < \angle ACB$. Assume that $\angle DBC < \angle ACB$. With C as a vertex, let $\angle ECB$ be the angle, as drawn in Fig. 7–3, that is congruent to $\angle DBC$. (Why is there such an angle?) Then the ray CE is in the interior of $\angle ACB$. (Why?) Hence the segment AB contains a point F in the ray CE. (This is obvious from the figure, and it may be proved with what we know about betweenness and interior of angles.) Now consider $\triangle FCB$. $\angle DBC$ is an exterior angle of this triangle which is congruent to the opposite interior angle FCB. This has already been shown to be impossible. Hence we cannot have $\angle DBC < \angle ACB$. The only possibility then is $\angle DBC > \angle ACB$. This completes the proof.

Second Proof. Using the same labeling for the triangle, let G be the point on BC, such that $BG \cong GC$. Let H be the point on the line AG that is on the other side of G from A, such that $AG \cong GH$. Then $\triangle GCA \cong \triangle GBH$. (Why?) Hence $\angle C \cong \angle GBH$. The ray BH is interior to $\angle DBC$. (This is obvious from Fig. 7–4, but it needs proof. It is interesting to note that Euclid neglects to prove this. See Exercise 3 below.) Therefore, $\angle DBC > \angle HBC$, and so $\angle DBC > \angle C$. (Why?)

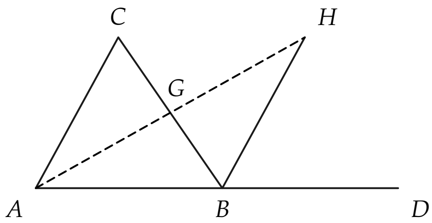


FIGURE 7–4

Remark. In Chapter 8, when we study parallel lines, we shall be able to prove that an exterior angle is congruent to the "sum" of the opposite interior angles, and from this fact, Theorem 7–1 follows easily. What is interesting is that Theorem 7–1 may be proved without any knowledge of properties of parallel lines.

Exercise Group 7–2

1. What kind of angles are the exterior angles of a right triangle?

2. Use Theorem 7–1 to prove that the angles opposite the legs of a right triangle are acute angles.

*3. Prove that the ray BH is interior to $\angle DBC$. [*Hint:* first show that H is on the same side of line BC as D. Since G is on the same side of BD as C, if H were on the opposite side of BD from C, then the segment GH would contain a point of the line BD. This is absurd.]

4. Prove the following corollary to Theorem 7–1. There is at most one line perpendicular to a given line l from a given point P. [*Hint:*

Suppose that there were two perpendiculars, as in Fig. 7–5.]

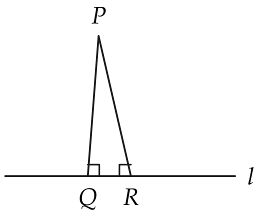


FIGURE 7–5

7–3 INEQUALITIES IN TRIANGLES

In this section we prove some theorems about the relative sizes of sides and angles in a triangle. We arrive at a proof of a theorem that, in effect, says that a line segment is the shortest path between two points.

Theorem 7–2. *If, in* $\triangle ABC$, $AB > AC$, *then* $\angle C > \angle B$, *and conversely, if* $\angle C > \angle B$, *then* $AB > AC$.

Remark. The meaning of this theorem can be stated briefly as "the greater side is opposite the greater angle."

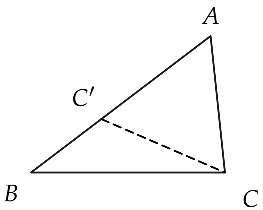


FIGURE 7–6

Proof. To prove the first part of the theorem, suppose that $AB > AC$, as shown in Fig. 7–6. Choose C' on AB, such that $AC' \cong AC$. Then by Theorem 7–1, $\angle AC'C > \angle B$. But $\angle AC'C < \angle C$. Hence $\angle C > \angle B$.

The converse follows readily from the fact that any two segments, are either congruent to each other or one is greater than the other. The argument is given below.

Given that $\angle C > \angle B$, we wish to show that $AB > AC$. Exactly one of the relations $AB < AC$, $AB \cong AC$, or $AB > AC$ is true. If $AB \cong AC$, then we would have $\angle C \cong \angle B$, which contradicts the given fact that $\angle C > \angle B$. If $AB < AC$, then by the first part of the theorem, we would have $\angle C < \angle B$, which also contradicts the given fact that $\angle C > \angle B$. Hence the only possibility is that $AB > AC$.

Exercise Group 7–3

1. Which angle in Fig. 7–7 is the smallest? Which is the largest? The numbers are lengths of the sides.
2. If in $\triangle EFG$, $\angle E$ has a measure of 61°, and $\angle F$ a measure of 54°, which side is the longer, FG or EG? Can you say anything yet about the length of EF?
3. In $\triangle PQR$, S is a point of PR, such that QS bisects $\angle PQR$ (see Fig. 7–8). Prove that $PQ > PS$. [*Hint:* Prove first that $\angle PSQ > \angle PQS$.]

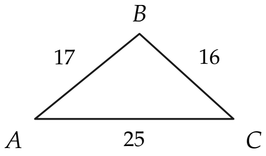


FIGURE 7–7

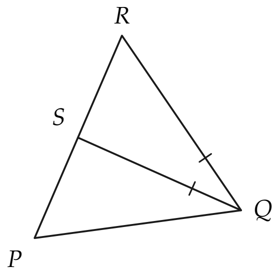


FIGURE 7–8

We now come to a very important and familiar inequality for triangles, called the *triangle inequality*.

Theorem 7–3. *In any* $\triangle ABC$, $\overline{AB} + \overline{BC} > \overline{AC}$.

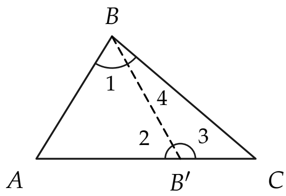


FIGURE 7–9

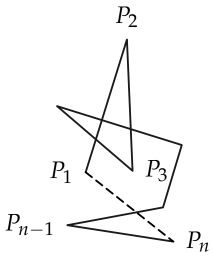


FIGURE 7–10

Proof. If AC is not the longest of the sides, the inequality is obvious. Therefore we may confine our attention to the case in which $\overline{AC} > \overline{AB}$ and $\overline{AC} > \overline{BC}$, as in Fig. 7–9.

STATEMENTS	REASONS
1. There is a point B' on AC, such that $AB \cong AB'$.	Why?
2. $\angle 1 \cong \angle 2$.	Why?
3. $\angle 3 > \angle 1$.	Why?
4. $\angle 2 > \angle 4$.	Why?
5. $\angle 3 > \angle 4$.	This follows from Statements 2, 3, and 4. Why?
6. $\overline{B'C} < \overline{BC}$.	Why?
7. $\overline{AB'} + \overline{B'C} < \overline{AB} + \overline{BC}$.	Statements 1 and 6 and properties of numbers.
8. $\overline{AC} < \overline{AB} + \overline{BC}$.	$\overline{AC} = \overline{AB'} + \overline{B'C}$.

Remark. This theorem is sometimes loosely stated as "a straight line is the shortest distance between two points." In fact, this last statement is sometimes used as a postulate, although incorrectly, since one must first have a notion of distance or of length of a segment.

Having proved the triangle inequality, we now see that the lines we have been talking about all along are, after all, "straight" in the precise sense of the triangle inequality. Perhaps this can be made clearer by defining *polygonal path* and *length of a polygonal path*.

Definition 7–2. Let P_1, $P_2, \ldots, P_n$ be any set of n points (Fig. 7–10). The set of points on all the segments P_1P_2, $P_2P_3, \ldots, P_{n-1}P_n$ is called a *polygonal path*. The path is said to join, or connect, the *endpoints* P_1 and P_n. The points $P_1, \ldots, P_n$ are called the *vertices*. In particular, if $P_1 = P_n$, the polygonal path is called a *polygon* and designated as polygon $P_1P_2 \ldots P_{n-1}$.

Definition 7–3. The *length of the polygonal path* $P_1 \ldots P_n$ is defined to be $\overline{P_1P_2} + \overline{P_2P_3} + \cdots + \overline{P_{n-1}P_n}$.

Corollary 7–3–1. *The length of a polygonal path is greater than, or equal to, the length of the segment joining its ends.*

Exercise Group 7–4

1. Two sides of a triangle have lengths of 5 and 7 in. What can you say about the third side?

2. Can the numbers 6, 8, and 15 be the lengths of the three sides of a triangle?

3. Prove that the difference between the lengths of two sides of a triangle is less than the length of the third side.

*4. Use the triangle inequality and Fig. 7–11 to show that if $\angle C > \angle B$, then $AB > AC$.

*5. In Fig. 7–12, $AC \cong A'C'$, $BC \cong B'C'$, and $\angle ACB > \angle A'C'B'$. Prove that $\overline{AB} > \overline{A'B'}$. [*Hint:* Show how to find a point Q on AB, such that $QB \cong QB'$, and use $\overline{AB} = \overline{AQ} + \overline{QB} = \overline{AQ} + \overline{QB'}$.]

*6. Use the result of Exercise 5 to prove that if, in two triangles, $\triangle ABC$ and $\triangle A'B'C'$, $AC \cong A'C'$, $BC \cong B'C'$, and $AB > A'B'$, then $\angle C > \angle C'$. Show that the statements $\angle C \cong \angle C'$ and $\angle C < \angle C'$ are false.

*7. If D in Fig. 7–13 is a point inside $\triangle ABC$, show that $\overline{AD} + \overline{DC} < \overline{AB} + \overline{BC}$. [*Hint:* Consider first $\triangle ABE$ and $\triangle DEC$.] Show also that $\angle ADC > \angle ABC$.

8. By using Theorem 7–2, prove that an equiangular triangle is equilateral.

9. If, in Fig. 7–14, $l \perp m$, and $l' \perp m$, prove that l and l' do not intersect. [*Hint:* Suppose they do.]

10. Show that a polygonal path connecting points A and B could have four vertices and still have the same length as AB.

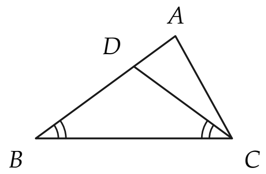


FIGURE 7–11

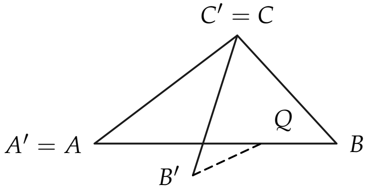


FIGURE 7–12

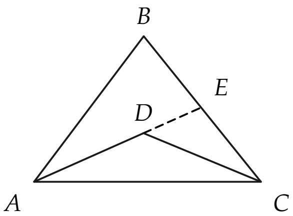


FIGURE 7–13

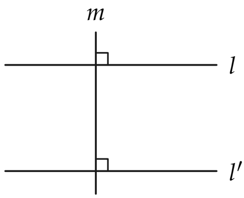


FIGURE 7–14

7–4 CIRCLES

There are two ways that we could define a circle. One is to use the idea of congruent segments, and the other is to use the idea of distance. Because two segments are congruent if and only if they have the same length, these two ways are equivalent.

Definition 7–4. A circle is determined by a point O, called the *center*, and a positive number r. The *circle* is the set of all points A of the plane, such that $\overline{OA} = r$. If A is a point of the circle, then the segment OA is called a *radius* (plural, *radii*). If A and B are on the circle, and the segment AB contains O, then AB is called a *diameter*.

Remark 1. You will note that we have not said that a circle is a "curve," because nowhere in our geometry have we defined what "curve" might mean. The circle is a set of points, and in the definition, we have told only how to decide whether any point A is, or is not, a point of the circle.

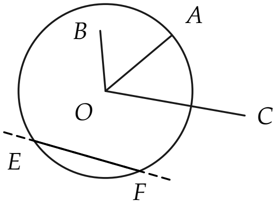
Remark 2. According to the definition, a radius is a segment, that is, a set of points. It is common usage to also call the real number r the radius. We will also use this language, so that the circle can be described as the "circle with center O and radius r." Similarly, if AB is a diameter, the length $\overline{AB} = 2r$ will be called the diameter. No confusion will arise, because it will always be clear whether we are talking about a set of points or a number.

Remark 3. We have not mentioned the inside or outside of the circle as yet, because these sets have not yet been defined.

Definition 7–5. If a circle (Fig. 7–15) has center O and radius r, the *interior* of the circle is the set of all points B, such that $\overline{OB} < r$. The *exterior* of the circle is the set of all points C, such that $\overline{OC} > r$.

A *chord* is a segment determined by two points E and F on the circle; the line through E and F is a *secant*.

A *central angle* of a circle is an angle whose vertex is at the center of the circle. An *inscribed angle* (such as $\angle EFG$) is an angle whose vertex is on the circle, and having points of the circle on each of the two sides of the angle.



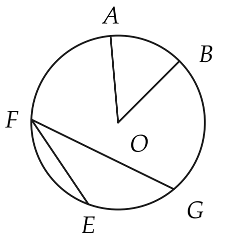


FIGURE 7–15

Exercise Group 7–5

1. A circle has center A and radius 5. The points B, C, D, E, are such that $\overline{AB} = 4$, $\overline{AC} = 5$, $\overline{AD} = 17$, and $\overline{AE} = 5$. Where are these points? That is, in what sets are they?

2. If AB is a diameter of a circle with center C, what is $\angle ACB$?

3. In Fig. 7–16, the circle has center O. What kind of angles are $\angle A$, $\angle B$, $\angle C$, $\angle D$, and $\angle E$?

4. Give a definition of circle in terms of congruence instead of distance.

5. If A is a point of a circle and O is the center, what can be said about a point B, such that $\overline{OB} \leqq \overline{OA}$?

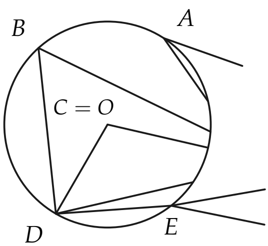


FIGURE 7–16

7–5 RULER AND COMPASS

In the preceding section we defined a circle in terms of distance. When drawing a circle on a piece of paper, we use the fact, familiar from experience, that the distance between the two points of the compass stays fixed. In this section we define those drawings that are called "ruler and compass constructions." Do not think that these are the only constructions possible. They are only the simplest kind. The ruler used in these constructions is just an unmarked straightedge.

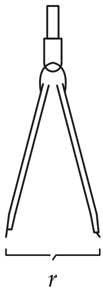


FIGURE 7–17

Definition 7–6. To make a *ruler and compass* (RC) *construction* is to draw a picture of a set of points, using only a ruler and compass. The ruler can only be used to draw a line through two points. The compass can only be used to draw a circle with a given center and radius.

Remark 1. RC constructions are sets that can be illustrated with certain mechanical devices. Circles and lines exist in our geometry independently of these mechanical devices. Interest in RC constructions originates in the simplicity of the tools needed to make them.

Remark 2. To draw a line or circle in an RC construction, two points are needed. We may not slide the ruler, mark it, or guess the opening of the compass. We must have two points already constructed in order to draw a line or circle.

Remark 3. The study of RC constructions at this point is not really necessary for the logical development of our geometry, but it does help us to visualize geometric figures, and it demonstrates the usefulness of what has already been studied.

Remark 4. It will be clear that the knowledge of RC constructions is quite useful to a draftsman.

7–6 HISTORICAL NOTE

The ancient Greeks were much attracted by RC constructions and attempted to solve all sorts of problems with these two tools. However, they encountered several problems that they could not handle using only ruler and compass. This does not mean that these problems are not solvable—only that a ruler and compass are not enough. Three of the problems that puzzled the Greeks have had a long and interesting history, which we cannot pursue here. They are:

(a) *The trisection of an angle.* Given any $\angle A$, the problem is to construct angles B, C, and D, as in Fig. 7–18, such that $\angle B \cong \angle C \cong \angle D$.

(b) *The duplication of a cube.* Suppose that a cube has side of length x. The problem is to construct the side of a cube having twice the volume of the given cube.

(c) *Squaring the circle.* Given a circle, the problem is to construct a square with the same area as the circle (Fig. 7–19).

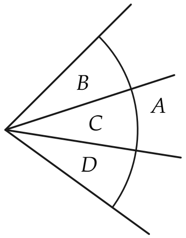


FIGURE 7–18

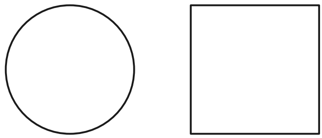


FIGURE 7–19

All these problems have been completely resolved in modern times, and *all are impossible of solution with ruler and compass alone*. This is a fact beyond dispute. It does not mean that in the trisection problem, for example, angles B, C, and D do not exist. It means that one cannot draw them using ruler and compass alone. When you think about this a moment, it may not be so surprising. No one is surprised that it is impossible to build a large building with only a tack hammer. A ruler and compass alone are like a tack hammer in so far as these problems are concerned.

There are, however, many constructions possible with ruler and compass. We will study some of them.

7–7 COPYING ANGLES AND TRIANGLES

$\angle A$ and a ray from point B are given, as in Fig. 7–20. We wish to construct an angle at B congruent to $\angle A$. Of course, from the existence postulate for angles, there are two such angles, as shown by the dotted lines in the figure. When we draw the angle at B congruent to $\angle A$, we say that we have "copied" $\angle A$.

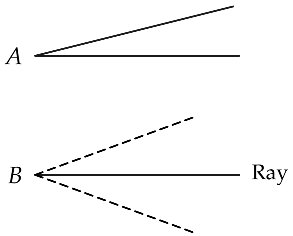


FIGURE 7–20

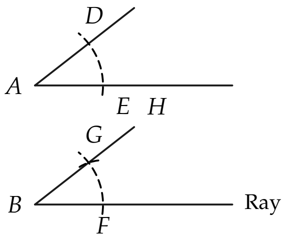


FIGURE 7–21

Construction 7–1. Draw a circle of radius r, with center A. This circle will intersect the sides of $\angle A$ in two points D and E (Fig. 7–21). (Why?) Draw a circle of the same radius r with center B. This circle will intersect the ray from B in a point F. Draw a circle, with center F and radius $\overline{DE}$. On a given side of the line BF, the two circles will intersect in a point G. $\angle GBF$ is congruent to $\angle A$.

You should recognize that there are two things to prove. Because the entire construction is centered around finding a point in which our circles intersect, we must first prove that such a point exists in our geometry. Of course, when we draw the picture, it is clear that there should be such a point. The real question, however, is whether we can take the assumptions we have made and prove that there is such a point. After we establish the existence of this point, we must prove further that the angles are congruent to each other. These two proofs should be considered separately.

Proof. (a) (The existence of a point of intersection of the circles.) By the existence postulate for angles, there is exactly one ray BH, on a given side of BF, such that $\angle FBH \cong \angle A$. The existence of the *ray* is thus assured. The existence of the points D, E, and F follows from theorems of measurement and congruence. There is a point G on the ray BH, such that $BG \cong BF \cong AE \cong AD$. Since $\angle A \cong \angle B$, $\triangle ADE \cong \triangle BGF$ by the S.A.S. theorem. Therefore, $GF \cong DE$. Hence, the point G is on the circle with radius $\overline{DE}$ and center F. Since G is also on the circle with center B and radius $\overline{BF}$, it is therefore on both circles and is a point of intersection of the two circles.

Proof. (b) (The congruence of the angles.) We now consider any point G on both circles. $AD \cong AE \cong BG \cong BF$. Since both circles have the same radius, $DE \cong GF$, and $\triangle AED \cong \triangle BFG$ by the S.S.S. theorem. Hence $\angle A \cong \angle B$.

In a similar manner we can copy a triangle.

Construction 7–2. Given $\triangle ABC$ (Fig. 7–22) and a ray from a point A', use the previous construction to copy $\angle A$. Then draw a circle, with center A' and radius $\overline{AB}$, which intersects the other side of $\angle A'$ at B'. Then draw a circle, with center A' and radius $\overline{AC}$, which intersects the given ray in C'. Then $\triangle ABC \cong \triangle A'B'C'$.

Proof. The proof is left to the student.

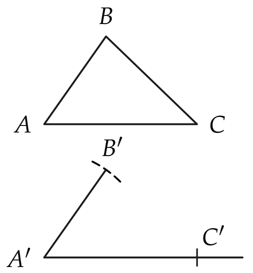


FIGURE 7–22

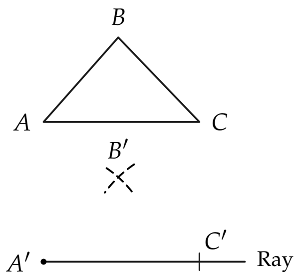


FIGURE 7–23

Construction 7–3. Given $\triangle ABC$ (Fig. 7–23) and a ray from a point A', draw a circle, with center A' and radius $\overline{AC}$, which intersects the ray in C'. Then draw circles, with centers at A' and C' and radii $\overline{AB}$ and $\overline{BC}$, respectively. These two circles will intersect, on a given side of the ray, in exactly one point B'. We will then have $\triangle ABC \cong A'B'C'$.

Proof. The proof is left to the student. Remember, you must first prove that B' exists and then prove $\triangle ABC \cong \triangle A'B'C'$.

7–8 OTHER CONSTRUCTIONS

It should now be clear that proving the validity of certain constructions involves first the proof of the existence of certain points and lines, and secondly the proof that the construction produces the desired result. In the remainder of our work on constructions, we will concentrate our efforts on the second part. The proof of existence of certain points can actually be quite hard at times. For example, when we try to prove that the construction for the perpendicular bisector of a segment is valid, we need a theorem, Theorem 7–4, that is very difficult to prove at this time.

Theorem 7–4. *If AB is a segment, and circles are drawn, with centers at A and B and radius $r > \frac{1}{2}\overline{AB}$, then these circles intersect in exactly two points P and Q, on opposite sides of the line through A and B (Fig. 7–24).*

Remark. From what we know about measurement of length, we know that every line segment AB has exactly one midpoint C, that is, a point C such that $\overline{AC} = \overline{BC} = \frac{1}{2}\overline{AB}$. Likewise, it is easy to show that every angle has a bisector, that is, from the vertex of the angle, there is a ray that forms, with the sides of the given angle, two congruent angles.

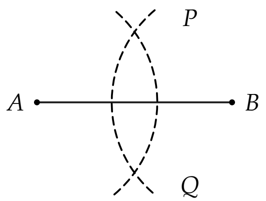


FIGURE 7–24

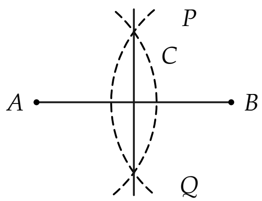


FIGURE 7–25

The constructions of this section show how to make RC constructions for midpoints of segments and bisectors of angles; but the *existence* of these midpoints and bisectors has been obtained independently of these constructions.

Construction 7–4. Draw circles, with centers A and B and equal radii greater than $\frac{1}{2}\overline{AB}$. These circles intersect at points P and Q. The line PQ is the desired perpendicular bisector (Fig. 7–25).

Proof.

STATEMENTS	REASONS
1. $AP \cong PB \cong AQ \cong QB$.	Construction.
2. $\triangle APQ \cong \triangle BPQ$.	S.S.S. theorem.
3. PQ intersects line AB in C.	P and Q are on opposite sides of AB.
4. $\triangle APC \cong \triangle BPC$.	S.A.S. theorem.
5. $AC \cong CB$, and $\angle ACP \cong \angle BCP$.	Statement 4.
6. PQ is the $\perp$ bis AB.	Statement 5.

Exercise Group 7–6

1. Prove that each segment has exactly one perpendicular bisector.
2. Prove that every point on the perpendicular bisector of AB is equidistant from A and B.
3. Prove that if a point R is equidistant from A and B, then it lies on the perpendicular bisector of AB.
4. Construct the perpendicular bisector of a segment by constructing two points on the same side of the line and equidistant from the ends of the segment.

Construction 7–5. To construct the perpendicular to a line at a point P on the line (Fig. 7–26), draw a circle, with center P and arbitrary radius, intersecting the line in points A and B. Then construct the perpendicular bisector of AB.

Construction 7–6. To construct a perpendicular to a line from a point not on the line, choose a point A on the line, and draw the circle with center P and radius AP (Fig. 7–27). The circle will intersect the line in a second point B (if A is properly chosen). Then construct the perpendicular bisector of AB, which passes through P.

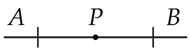


FIGURE 7–26

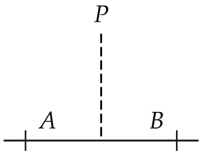


FIGURE 7–27

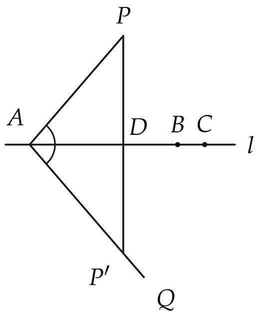


FIGURE 7–28

Proof. We first show that there actually is a line through P perpendicular to the line. Then we show that the RC construction produces this line.

First choose any point A on l and any other point C on l (Fig. 7–28). Let AQ be the ray, on the opposite side of l from P, such that $\angle PAC \cong \angle QAC$. (There is only one such ray, by the existence postulate for angles.) Choose P' on AQ, such that $AP' \cong AP$. Then PP' intersects l in a point D. (Why?) If $A \neq D$, then triangles are formed, and by S.A.S., $\triangle APD \cong \triangle AP'D$. Therefore, $\angle ADP$ and $\angle ADP'$ are right angles, and $PP' \perp l$. If

$A = D$, there are no triangles, but then, because of the way AQ was selected, $PP' \perp l$. (Why?)

Having established that a perpendicular from P exists, we now show that the RC construction gives us this perpendicular. If $D \neq A$, there is a point B on the opposite side of D from A, so that $AD \cong BD$. It follows that the circle with center P and radius AP does intersect the line in a second point B, and this was the main part to be proved in the construction. Now the perpendicular bisector of AB can be drawn, and P lies on it. (Why?)

Exercise Group 7–7

1. Construct a right triangle, having been given the lengths of the two legs.

2. Construct points D, E, and F on the segment AB, such that $\overline{AD} = \overline{DE} = \overline{EF} = \overline{FB} = \frac{1}{4}\overline{AB}$.

*3. Suppose that P is not on a line l, and that a given circle with center at P intersects l in exactly one point Q. Prove that PQ is perpendicular to l. [*Hint:* If PQ were not perpendicular, then the construction in the proof of Construction 7–6 would lead to a contradiction.]

4. Construct the perpendicular from P by choosing A, as in Fig. 7–29, constructing $\angle 2 \cong \angle 1$, and point P', such that $AP \cong AP'$.

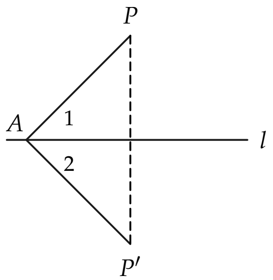


FIGURE 7–29

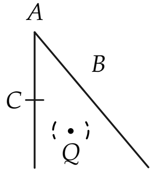


FIGURE 7–30

Construction 7–7. To bisect $\angle A$, draw a circle, with center A, intersecting the sides of $\angle A$ in points B and C (Fig. 7–30). Then construct a point Q on the perpendicular bisector of BC and in the interior of $\angle A$. The ray AQ bisects $\angle A$.

Proof. The point Q can be found by a previous construction. $\triangle ABQ \cong \triangle ACQ$, by S.S.S. Hence $\angle BAQ \cong \angle CAQ$.

Exercise Group 7–8

1. Draw an acute angle, and construct its bisector.

2. Construct and bisect a right angle.

3. Bisect an obtuse angle.

4. Draw a triangle, and construct the bisectors of its angles. (They should meet in one point.)

5. If P is on the bisector of $\angle A$, and B and C are points on the sides of $\angle A$, such that $AB = AC$, prove that $PB \cong PC$.

6. Prove that an angle has only one bisector.

7. Suppose that B and C are points on the two sides of $\angle A$, such that $AB \cong AC$. Show that if M is the midpoint of BC, then AM bi-

sects $\angle A$.

*8. Figure 7–31 shows another construction for the angle bisector. Consider two circles, with center A and radii r and r', such that $r' > r$. These circles intersect the sides of A in four points C, C', D, and D'. The lines through these points intersect in a point P on the bisector. (Why?)

*9. If, in Fig. 7–32, $\angle CAD \cong \angle EBF$, $\angle 1 \cong \angle 2$, and $\angle 3 \cong \angle 4$, then $\angle 2 \cong \angle 4$. *Note:* This theorem is often stated as "halves of congruent angles are congruent to each other." Prove this theorem.

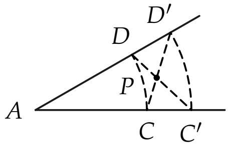


FIGURE 7–31

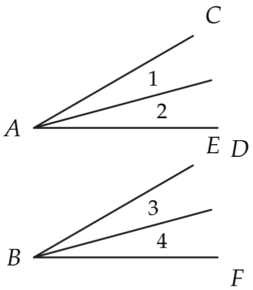


FIGURE 7–32

REVIEW OF CHAPTER 7

In this chapter we have used the congruence theorems to study inequalities in triangles and verify constructions.

(a) Make a list of all the theorems on inequalities in triangles, then study their proofs.

(b) Make a list of the different RC constructions. Prove them correct without looking at the book.

Review the following definitions, which you met for the first time in this chapter: exterior angle, circle, interior of circle, exterior of circle, radius, diameter, chord, secant, central angle, inscribed angle, RC construction.

Review Exercises

1. Construct an angle of 45°.
2. Construct an angle of 60°, of 30°.
3. Construct an angle of 15°, of 75°.
4. Construct an equilateral triangle.
5. Construct a right triangle having acute angles of 60° and 30°.
6. Draw an obtuse triangle, and construct the bisectors of its angles.
7. Draw an angle. Copy it, using ruler and compass.
8. Draw three separate segments. Construct a triangle having sides congruent to the given segments.
9. Draw a line, and mark a point P not on the line. Construct the perpendicular to the line, passing through P. How do you know that there is only this one perpendicular?
10. Draw a triangle. Construct the altitudes of the triangle.
11. Draw a triangle. Construct the perpendicular bisectors of its sides.
12. Construct an isosceles triangle if the vertex angle and the congruent sides are given.
13. Draw a segment. Construct a circle having this segment as a diameter.
14. In a right triangle, which side is longest? Why?
15. Prove that the sum of the lengths of three sides of a quadrilateral is greater than the length of the fourth side.
16. $\angle ABC$ is a right angle, and

C is between B and D. Prove that $AD > AC$. (Fig. 7–33)

17. Prove that $\overline{AD} + \overline{CB} > \overline{AB} + \overline{CD}$. (Fig. 7–34)

18. AE bisects $\angle DAF$, and AC bisects $\angle DAB$. (Fig. 7–35) Prove that $\angle CAE$ is right. [*Hint:* Extend AE.]

19. AC bisects $\angle DAB$, and $\angle CAE$ is right. Prove that AE bisects $\angle DAF$. (Fig. 7–36)

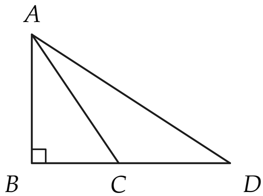


FIGURE 7–33

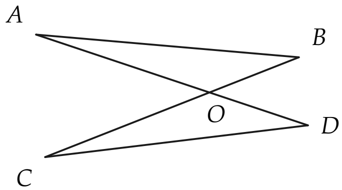


FIGURE 7–34

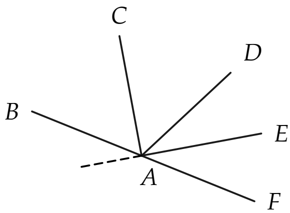


FIGURE 7–35

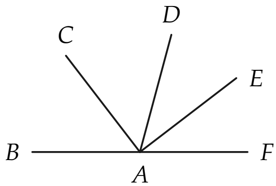


FIGURE 7–36

ALGEBRA REVIEW

1. In $\triangle ABC$, $AB < BC < CA$. The greatest angle in this triangle has a measure of 32° more than the measure of the smallest angle. The sum of the measures

of these two angles is $120°$. Determine the number of degrees in the smallest angle in this triangle.

2. An exterior angle at A of $\triangle ABC$ has a measure of $65°$. The sum of the measures of angles A and B is $145°$, and the sum of the measures of angles A and C is $150°$. Determine the number of degrees in each angle of the triangle.

3. Points O, A, and B are collinear. $\overline{OA}$ is 7 in. more than $\overline{BA}$. A is not between O and B. Determine the length of OB.

4. $\overline{AB} = 17$, and $\overline{AC} = 12$. The number $\overline{BC}$ is a root of the equation $(x-6)(x-4) = 0$. Determine $\overline{BC}$.

5. If B is between A and C, $\overline{AC}/\overline{BC} = \frac{5}{3}$, and $\overline{AB} = 10$ in., determine the lengths of AC and BC.

6. B is between A and C, and $\overline{AB}/\overline{AC} = \frac{3}{5}$. The midpoint of AC is M, and $\overline{MB} = 3$. Determine $\overline{AC}$.

*7. A, B, C and D are points in this order on a line. The length of AC is twice the length of BD. The length of AB is four times the length of CD. The length of BC is 4. Determine the lengths of AB and AD.

*8. The difference of the squares of two numbers is 33. The sum of these numbers is 11. Determine the numbers.

*9. Analyze the following mathematical trick. Choose any number; multiply this number by 2; add 4 to this product; multiply by 5; divide by 10; subtract from this result the original number. The result will be 2, no matter what number was originally chosen.